Coach space: alternatives and selection

A consistent geometry basis for procurement discussion and engineering review

Option	Cabin	Clear width	Regular seating	Places	Aisle	Assessment
A	24 ft	90 in	Longitudinal benches	24	50 / ~30 in*	Compact / lower capacity
B	24 ft	104 in	7 rows, 2 + 2	28	24 in	Shorter transverse option
C	**26 ft**	**104 in**	**8 rows, 2 + 2**	**32**	**24 in**	**Preferred for next review**
D	28 ft	104 in	9 rows, 2 + 2	36	24 in	More capacity / more swept area

What the preferred regular coach uses

306 in cabin length = 31 in front seat-to-wall zone + 8 x 20 in seat envelopes + 7 x 12 in row gaps + 31 in rear seat-to-wall zone.

104 in clear width = 40 in two-adult bench + 24 in aisle + 40 in two-adult bench. Eight rows yield 32 fixed places.

The 20 in row envelope includes the back/cushion depth; 32 in pitch leaves 12 in between consecutive envelopes. This is not a measured knee-clearance result.

40 in for two adults provides 20 in nominal width per person. 29.5 in provisional occupied depth extends 9.5 in beyond each seat; verify knees, coats and circulation in a full-size mock-up.

Capacity is not yet approval

Aisle, vestibule, doorway, egress, occupant load and vehicle classification remain engineer / authority decisions. Do not copy the 24 in aisle into fabrication documents.

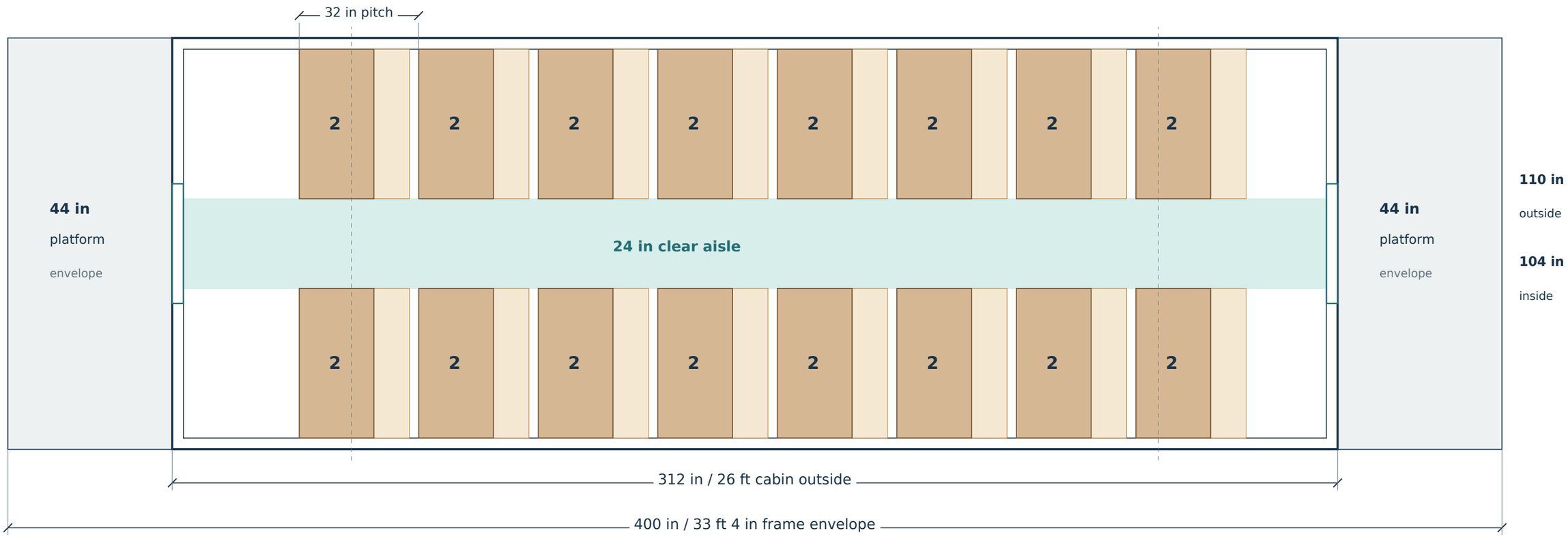
Three preferred regular coaches + one separate accessible coach = 96 regular places + 11 fixed accessible-car places + 2 occupied wheelchair bays = 109 total places.

28 ft option must earn its extra capacity through stability, axle load, vehicle sweep, traction and loading-cycle review. Its capacity advantage is not free.

*Alternative A: 50 in between 20 in seat envelopes; approximately 30 in occupied route after two 10 in leg reservations. A/B/D bodies are not modeled. C sums 26 ft 8 in bench length each side; D sums 30 ft.

Regular coach: preferred 32-place arrangement

C: 26 ft cabin | 104 in clear width | 8 rows | two adults per bench section



Geometry to test in the prototype

31 in seat-to-wall end zones. A 29.5 in seated-knee envelope leaves about 21.5 in behind the last occupied row. 32 in clear centred end doors; hardware and swings not selected.

End platforms add 44 in each. Gates, steps, guards, boarding equipment and station interfaces remain unresolved. Knee-space envelopes require a physical adult mock-up.

Numbers inside bench envelopes indicate adult places. Pale extensions are 9.5 in knee-projection tests. All 32 places are seated; no standing passenger capacity is claimed.

Interfaces remain conditional

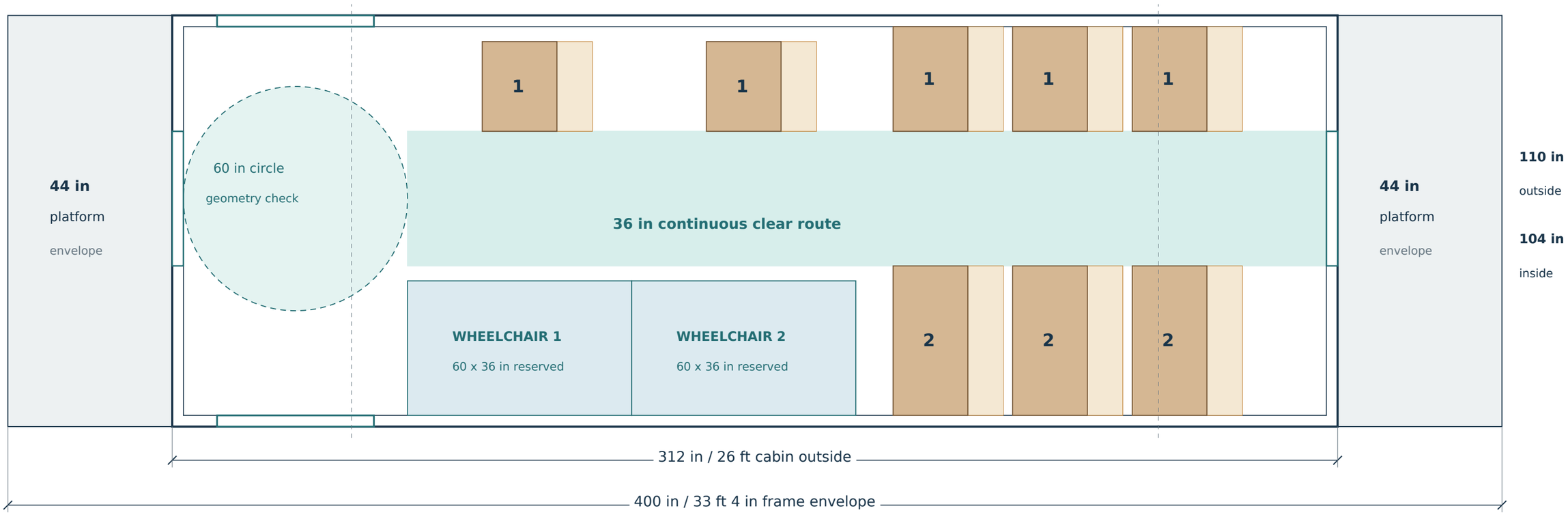
3 in wall buildup is a reserved envelope for structure, insulation, lining and services. Verify thermal and structural feasibility.

Candidate bogie centres = 216 in. This does not select wheelbase, wheel diameter, profile, bearings, springs or rated undercarriage.

Seats face the same direction for a compact row layout. Emergency exits, lighting, ventilation and HVAC duct routes require coordinated development.

Accessible coach: 11 fixed places + 2 wheelchair bays

Separate car | continuous 36 in route | 60 in front turning circle | conditional boarding interfaces



Reserved accessible space

Front zone x = 0-60 in is clear. 60 in turning circle centred at x = 30, y = 6. End doors share the route offset: y = -12 to +24 in.

Two bays each 60 x 36 in: x = 60-120 and 120-180, y = -52 to -16. A 4 in buffer separates the bays from the route at y = -12.

Two companion seats at x = 80-100 and 140-160, y = 24-48. Rear rows: 6 bench places plus 3 single places. Pale extensions are 9.5 in knee tests; rear occupied end zone about 22.5 in.

Boarding and evacuation are not yet resolved

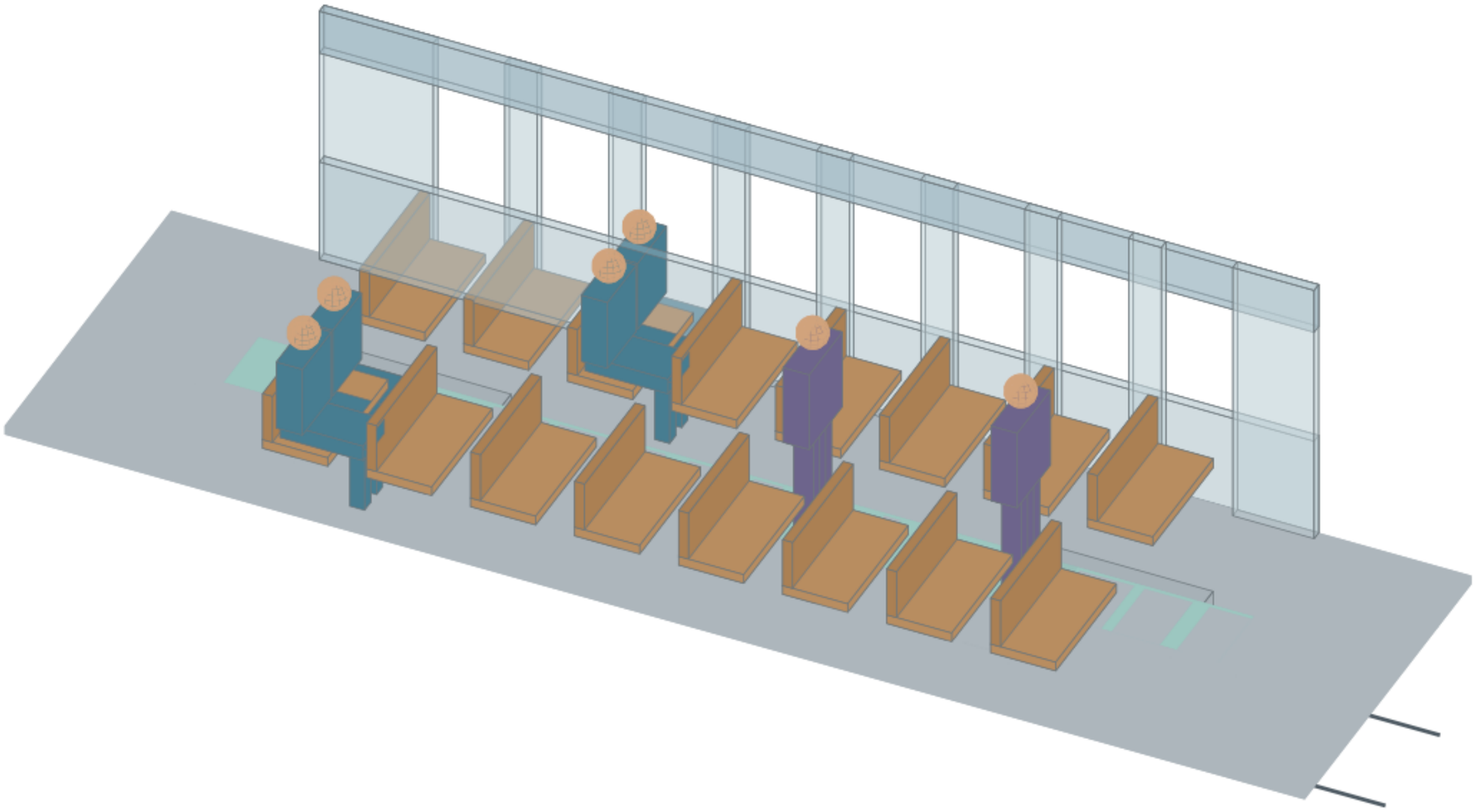
42 in clear side-door openings reserved on BOTH sides of the front zone, x = 9-51. Side boarding avoids relying on inaccessible end stairs.

Ramp / bridge plate or lift must be selected with the station level, gap, platform motion, 28 in provisional floor height and actual passenger devices.

Wheelchair orientation, securement if required, restraint interfaces, assisted evacuation, alternative escape and applicability of accessibility provisions need specialist review.

Regular coach: adult-scale cutaway

Space model generated from the same dimensions as the DXF and STEP general arrangement

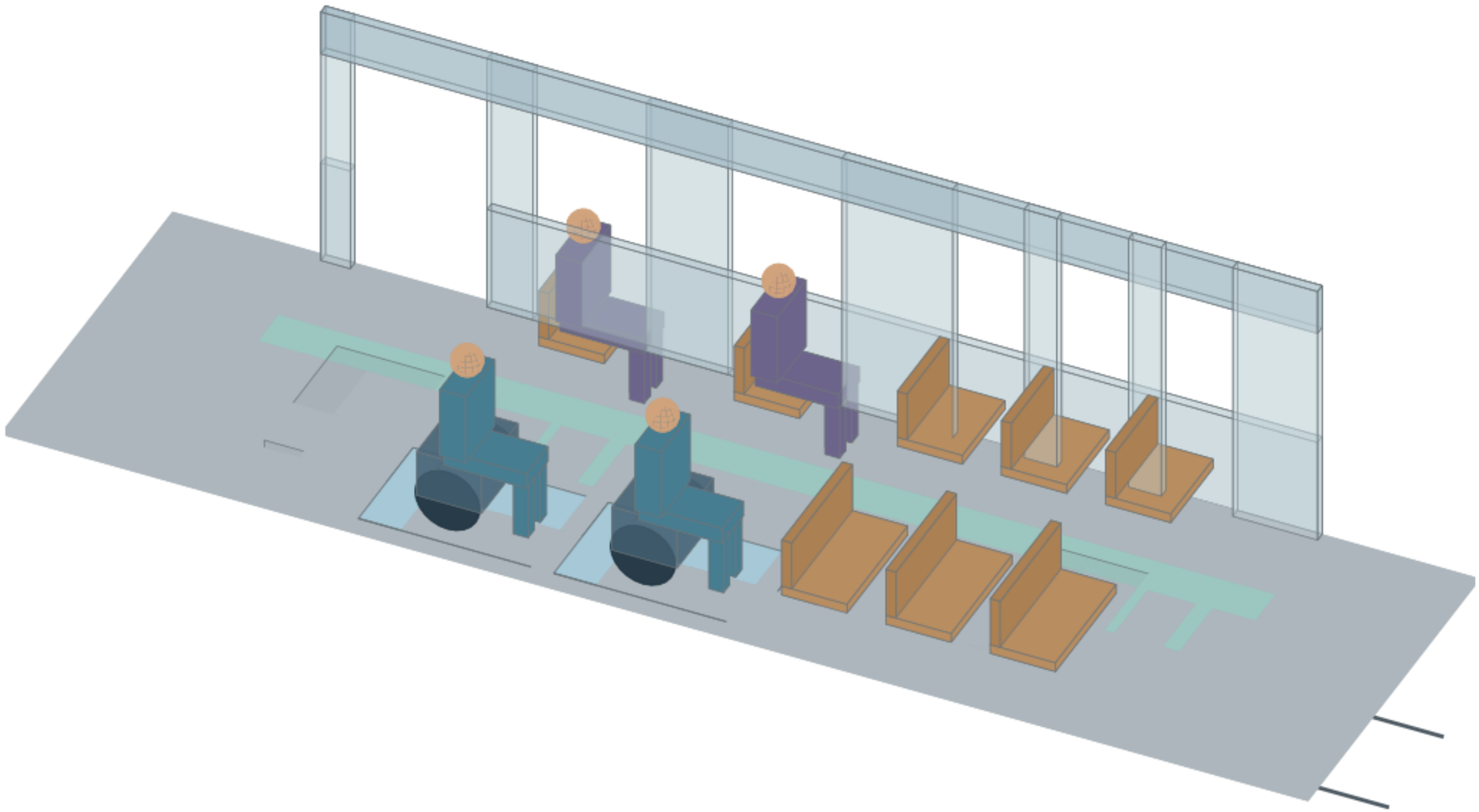


Six adult scale proxies: 4 seated (19 in shoulders) + 2 standing (22 in shoulders). Standing illustrates circulation only.

Roof / near wall omitted. Truck boxes are reserved packages. Adult comfort, knee clearance and occupied egress remain unverified.

Accessible coach: reserved-space cutaway

Separate accessible coach maintains dedicated boarding and circulation space

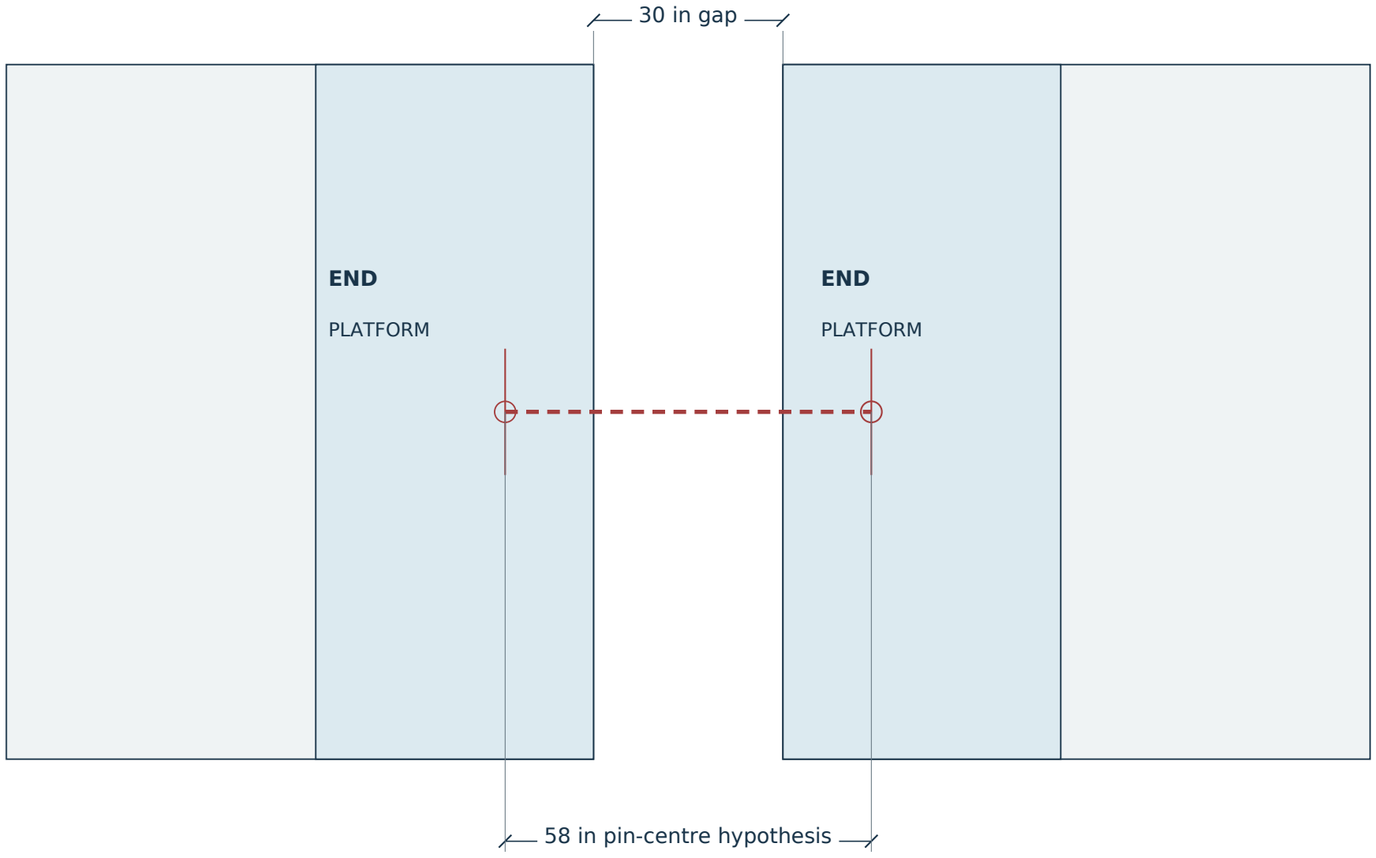


Two occupied wheelchair proxies and two seated companions illustrate allocation. Remaining fixed seats are unoccupied.

Roof / near wall omitted. Reserved bays are not a securement, lift, restraint or evacuation design; these decisions remain open.

Car connections and stairs: interface study

No coupler type, pin diameter, drawbar section, material, weld or rating is selected



Kinematic inputs only

14 in proposed reference-pin setback from each frame end; 30 + 14 + 14 = 58 in nominal pin-centre length.

Red dashed line is an axis, not a physical bar. Its thickness is not a component size.

Validate yaw, pitch, roll, curve compression, reverse curves, draft/buff forces and height mismatch before choosing hardware.

Retained connection, independent emergency response and protected hose/cable articulation require supplier / engineer design.

Onboard stairs and platforms

44 in end-platform envelope is reserved. Side stair locations and number of risers/treads are not frozen: they depend on floor height, station level, limits of the vehicle and boarding plan.

Prefer protected, gated boarding with slip-resistant treads, continuous handholds and a verified deployed / stowed envelope. Check all parts through curves and crossings.

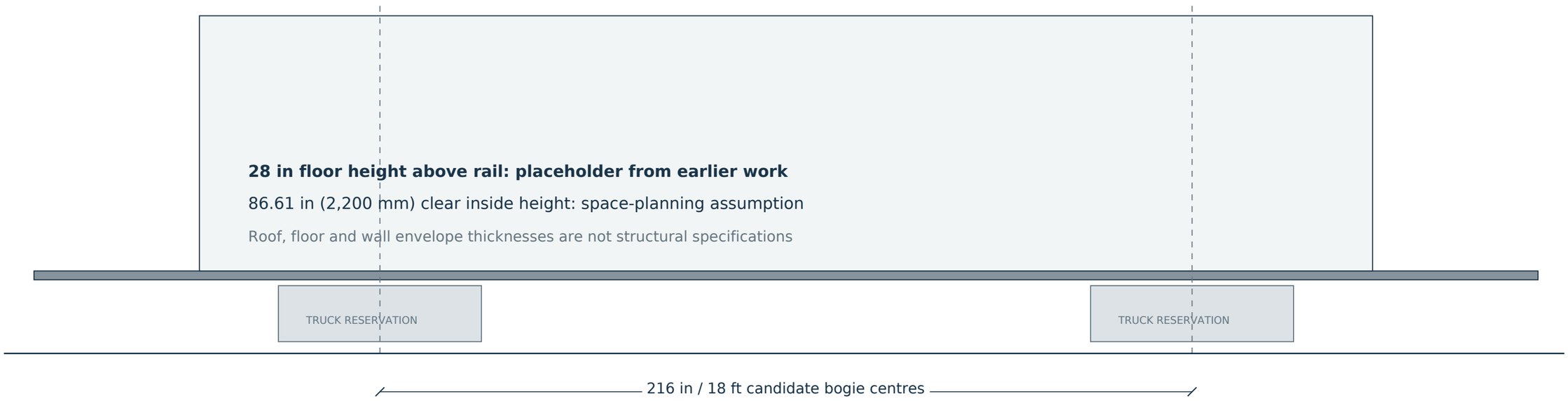
Protection and evidence required

No passenger circulation through a moving inter-car gap is assumed. Gates / barriers must prevent access to coupling and pinch zones.

Next drawing issue needs actual underframe attachments, coupler manufacturer ratings or calculated drawbar design, articulation limits, swept-envelope model and brake/control connectors.

Vertical envelope and coordination holds

All dimensions shown are preliminary reservations; structure and running gear are not designed



Release holds before detail engineering

H1: Surveyed profile, minimum radius and reverse-curve geometry -> wheel/rail, bogie centres, coupling motion and track clearances.

H2: Confirmed passenger design load, total weight and CG -> axle loads, stability, traction and braking.

H3: Vehicle classification, egress/accessibility decisions and actual equipment -> door/aisle/floor-height freeze.

H4: Certified component data and engineer calculations -> fabrication parts, materials, tolerances, welds and inspection criteria.

File meaning and checks

DXF: editable 2D general arrangement in inches. STEP: separate named solid envelopes in millimetres; no parametric feature tree is retained in STEP.

Python source + JSON preserve editable dimensions and regeneration logic. No fabrication drawings or certified load ratings are represented.

Validation reopens DXF / STEP, checks DXF audit and CAD solid validity, and verifies listed 2D seat/route separations. It does not certify all 3D clearances or passenger safety.

Body exterior and six-person views are geometric planning illustrations, not a selected historical finish, HVAC fit-out or locomotive design.